Adult Bradycardia

Appropriateness for clinical condition
Bradycardic patients requiring intervention generally have a heart rate less than 60bpm

Identify and Treat Underlying Cause
- Maintain patient airway, assist breathing as necessary
- Titrate oxygen to maintain Sp02 >94%
- Routine vital sign assessment
- 12-lead ECG (do not delay treatment for severe cases)
- Consider hypoxia and toxicological causes

Persistent bradycardia causing:
- Hypotension?
- Acutely altered mental status?
- Signs of shock?
- Ischemic Chest Discomfort?
- Acute Heart Failure?

NO

Consider Tele EMS

Monitor and observe

YES

Consider ALS Back-Up

Atropine
1mg IV
(repeat every 3-5 minutes) Max dose 3mg

EMT-A

Transcutaneous Pacing

AP

Consider Analgesia and Sedation
- Midazolam, or/
- Ketamine

EP

Dopamine Infusion	5-20**mcg**/kg/min IV	AP
Adrenaline Infusion	2-10**mcg**/min IV	AP
Adrenaline Push Dose	0.01**mg** IV (every 2-5 minutes)	AP

Patient Care Goals

1. Maintain adequate perfusion.
2. Identify and treat any underlying cause.
3. Appropriate management to improve end-organ perfusion.

Inclusion & Exclusion Criteria

Inclusion Criteria

- Adult patients with a Heart rate less than 60 beats per minute (BPM) with either symptoms (altered mental status (AMS), chest pain (CP), congestive heart failure (CHF), seizure, syncope, shock, pallor, diaphoresis) or evidence of hemodynamic instability.
- The common ECG rhythms classified as bradycardia include:
    - Sinus bradycardia
    - Second-degree AV block:
        - Type I-Wenckebach/Mobitz I
        - Type II-Mobitz II
    - Third-degree AV block, complete heart block
    - Ventricular escape rhythms

Exclusion Criteria

- Adult patients with a heart rate within normal ranges.
- Pediatric patients <13 years.

Key Patient Assessment Points

1. **12-Lead ECG**
2. **Investigation of possible non-cardiac causes of bradycardia**, such as:
    - Hypoxia
    - Shock
    - Second- or third-degree atrioventricular (AV) block
    - Toxin exposure (beta-blocker, calcium channel blocker, organophosphates, digoxin)
    - Electrolyte disorder
    - Hypoglycemia
    - Increased intracranial pressure (ICP)
3. **Evaluate for Signs of Instability,** including:
    - Hypotension
    - Acutely altered mental status
    - Signs of shock
    - Ischemic Chest Discomfort
    - Acute Heart Failure
4. **Determine Triage Category**
    - Determine Triage Category as per Patient Triage Categories CPG.

Patient Management & Interventions

1. **Primary Survey**
    - Manage airway as required, see Airway Management CPG.
2. **Administer oxygen**
    - Administer oxygen as appropriate with a target of achieving Sp02 >94%.
3. **Obtain ECG**
    - Initiate monitoring and perform 12-lead ECG
4. **Investigate alternative causes**
    - Check blood glucose and treat hypoglycemia, see Hypoglycemia CPG.
5. **If bradycardia and symptoms or hemodynamic instability continue:**
    - **Administer Atropine**

        - Atropine 1 mg IV/IO every 3–5 minutes (maximum total dose of 3 mg)
    - **Perform Transcutaneous Pacing**
        - Indicated if atropine is ineffective.
        - If pacing is performed, consider sedation and/or pain control:
            - Midazolam: 0.5-1mg IV/IO increments, every 5 minutes.
            - Ketamine: 0.25-1mg/kg IV/IO increments, every 3-5 minutes.
                - If larger doses of Ketamine are required, administer dose as an infusion in 100ml of Normal Saline over 5-10 minutes (to reduce the risk of adverse effects).

6. **Consider Vasopressors**
    - If the above interventions fail to increase heart rate, or if pacing is not available or not tolerated by the patient (no sedation available), consider:
        - Dopamine Infusion: 5-20mcg/kg/min IV/IO, or/
        - Adrenaline Push Dose: 0.01mg every 2-5 minutes, or/
        - Adrenaline Infusion: 2-10mcg/min IV/IO
7. **Patient Reassessment**
    - Frequently Reassess the patient as per Universal Care CPG.
    - Specific Patient Reassessment Points:
        - Electrical and Mechanical capture if Transcutaneous Pacing is used.

Patient Safety Considerations

- If pacing is performed sedation and pain control should be considered.

Clinical Insights & Additional Information

- Atropine Administration
    - Before administering atropine, rule out head injury, hypothermia, and hypoxia as causes of bradycardia; in these conditions, bradycardia may be a compensatory mechanism, and atropine administration could worsen the patient's condition by disrupting this protective response. Consider Telemedicine consultation for such cases.
    - Use caution with Atropine in myocardial infarction – increasing heart rate may increase myocardial oxygen consumption, worsening myocardial ischemia.
- Sedation and Analgesia
    - Administration of analgesia and sedation for pacing may be delayed for significantly compromised or patients with altered mental status.
    - Morphine may be considered for analgesia; however higher doses are generally required and may result in hypotension.
    - Morphine may also have a synergistic effect when administered with midazolam causing profound hypotension.
- Dopamine and Adrenaline Infusions
    - Dopamine or adrenaline may be considered when pacing is not effective or is not tolerated by the patient (if sedation is not available).
    - Dopamine or adrenaline may also be considered after successful pacing if the patient remains hypotensive.
    - Adrenaline may be easier to administer when a syringe driver is not available.

Documentation

- Identification: Full patient name, age, and gender.
- Mechanism / Medical complaint: Presenting symptoms (e.g., dizziness, syncope), initial rhythm, and suspected cause of bradycardia.
- Injuries / Information: Relevant history (e.g., medication, cardiac history, toxins, electrolyte imbalance).
- Signs:
    - Full vital signs including time-stamped pulse, blood pressure, respiratory rate, SpO2, GCS.
    - Two sets of vital signs unless critical patient condition precludes

    - ECG interpretation (rhythm classification: sinus bradycardia, Mobitz I/II, complete heart block).
    - Capture of any rhythm changes.
- Treatment:
    - Airway management steps and oxygen administration.
    - Blood glucose result and treatment if hypoglycemia.
    - All medications given: name, route, dose, time, and response.
    - Transcutaneous pacing details: start/stop time, rate, capture, analgesia/sedation provided.
    - Justification if pacing or medication was withheld.
    - If sedation used: dose, scale of agitation/pain (e.g., RASS), and reassessment.
- Allergies: Documented allergies and drug interactions.
- Medications: All pre-existing medications and any medications administered in the field.
- Background: Relevant medical history, comorbidities, and recent procedures (e.g., pacemaker).
- Other:
    - Communication with Telemedicine (time and instructions).
    - Response to interventions.
    - Consideration of reversible causes and actions taken.

Key Performance Indicators

1. Perform a 12-lead ECG on all patients with bradycardia.
2. Administer atropine or Transcutaneous Pacing for all hemodynamically unstable patients (where appropriate).

References

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